ABSTRACT

Solar energy is one of the most abundant and renewable sources of energy available on Earth. With the increasing demand for sustainable energy solutions, solar power systems have become an important alternative to conventional energy sources. However, the efficiency of solar panels is limited when they are fixed in a single position because the angle of sunlight changes throughout the day as the sun moves across the sky.This project presents the design and implementation of a dual-axis solar tracking system using Arduino. The main objective of this system is to maximize solar energy absorption by continuously aligning the solar panel toward the direction of maximum sunlight. The system uses four Light Dependent Resistors (LDRs) to detect sunlight intensity from four different directions. Based on the difference in light intensity detected by the sensors, the Arduino microcontroller controls two servo motors to adjust the position of the solar panel both horizontally and vertically.One servo motor controls the horizontal rotation (East-West direction) while the other controls the vertical tilt (North-South direction). The system compares the light intensity values from the sensors and moves the solar panel until balanced light intensity is achieved, indicating that the panel is directly facing the sun.The proposed solar tracking system is low cost, energy efficient, and easy to implement. It significantly improves solar panel performance compared to stationary solar panels. This project demonstrates how embedded systems and sensors can be effectively used to improve renewable energy systems.

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CHAPTER-1

INTRODUCTION

1.1 Introduction

Renewable energy sources play a crucial role in addressing global energy demands and reducing environmental pollution. Among the various renewable energy sources available today, solar energy is one of the most widely used and sustainable energy sources. Solar panels convert sunlight directly into electrical energy using photovoltaic cells. In conventional solar power systems, solar panels are usually fixed at a particular angle. While this setup is simple, it cannot capture maximum solar energy throughout the day because the position of the sun continuously changes from sunrise to sunset. As a result, a large portion of potential solar energy is lost. To overcome this limitation, solar tracking systems are used. A solar tracker is a device that automatically moves a solar panel to follow the sun's path across the sky. By maintaining the panel perpendicular to the sunlight, the amount of energy generated by the solar panel can be significantly increased.

Fig. 1.1.1: Solar Panel

Fig. 1.1.1 describes that Solar panels, fundamentally known as photovoltaic panels, operate on the profound principle of converting light energy directly into usable electrical energy. This process is driven by the photovoltaic effect, a physical and chemical phenomenon first observed in the mid-nineteenth century. At the heart of this technology lies the solar cell, typically constructed from semiconductor materials like silicon. When photons, which are particles of light from the sun, strike the surface of these semiconductor cells, they transfer their energy to electrons within the material. This sudden influx of energy excites the electrons, causing them to break free from their atomic bonds. The engineered structure of the solar cell, which features a meticulously designed electric field created by positive and negative layers, forces these liberated electrons to flow in a singular, defined direction. This continuous flow of electrons is what we recognize as a direct current, the foundational electrical output of a standard solar panel before it is processed for everyday household or commercial use.

Because the vast majority of modern residential and commercial electrical systems, as well as the wider municipal utility grids, operate on alternating current, the direct current generated by the solar panels cannot be used immediately. It must first pass through a critical component known as an inverter. The inverter seamlessly translates the direct current into alternating current, ensuring compatibility with standard household appliances and the broader electrical infrastructure. Once converted, this electricity can be routed directly into a building's electrical panel to offset real-time power consumption. If the solar array generates more electricity than is actively being consumed on the premises, the surplus power is often exported back to the local utility grid. In many regions, this export triggers a process called net metering, where the utility company credits the solar system owner for the excess energy, effectively turning the electrical meter backward and providing a tangible financial return on the initial investment.

The physical composition of the panels themselves heavily dictates both their overall efficiency and their visual characteristics. Monocrystalline panels are manufactured from a single, continuous crystal structure of silicon, which allows for an unimpeded flow of electrons, resulting in the highest efficiency rates and a distinctively uniform, dark aesthetic. Polycrystalline panels, alternatively, are created by melting together multiple fragments of silicon. While this manufacturing process is generally faster and more cost-effective, the borders between the individual silicon crystals create slight resistance for the moving electrons, rendering them slightly less efficient and giving them a signature blue, speckled appearance. Thin-film panels diverge entirely from traditional rigid silicon wafers, utilizing a process where photovoltaic materials are deposited in extremely thin layers onto flexible substrates such as plastic, glass, or metal. Although they require significantly more surface area to generate comparable amounts of electricity, their lightweight and adaptable nature makes them uniquely suited for unconventional applications where heavy, traditional solar panels would be structurally impractical.

1.2 Problem Statement:

The problem addressed in this project revolves around the need for improving the efficiency of solar energy generation systems. Solar energy is one of the most abundant and renewable sources of energy available on Earth, and it plays a crucial role in addressing the growing global demand for sustainable power generation.

However, conventional solar power systems commonly use fixed solar panels that remain stationary throughout the day. Because the position of the sun continuously changes from sunrise to sunset, the angle at which sunlight strikes the solar panel also changes.

In traditional solar installations, solar panels are mounted at a fixed orientation and tilt angle that is optimized for a particular time of the day or season. While this configuration allows the panel to receive adequate sunlight for a limited period, it cannot maintain optimal alignment with the sun throughout the day. During the early morning hours, sunlight falls on the panel at a shallow angle, resulting in reduced energy absorption. Similarly, in the afternoon and evening, the sun moves away from the panel's optimal position, again decreasing the efficiency of energy conversion. Since photovoltaic cells produce maximum electrical energy when sunlight strikes them perpendicularly, any deviation from this optimal angle leads to a significant loss of potential energy. Consequently, a large portion of available solar radiation remains unused in fixed solar panel systems.

Another challenge associated with conventional solar energy systems is the inability to adapt to changing environmental conditions. Factors such as seasonal changes, atmospheric conditions, and variations in sunlight intensity influence the amount of solar radiation received by a panel. Fixed solar panels cannot dynamically adjust their position in response to these variations, which further limits their efficiency. This problem becomes particularly important in regions where maximizing renewable energy generation is necessary to reduce dependence on fossil fuels and minimize environmental impact. Therefore, improving the efficiency of solar energy systems by ensuring continuous alignment with the sun has become an important research area in renewable energy technology.

To overcome these limitations, solar tracking systems have been proposed as an effective solution. Solar trackers are designed to automatically adjust the orientation of solar panels so that they continuously face the direction of maximum sunlight. By maintaining the solar panel perpendicular to the incoming sunlight, solar trackers can significantly increase the amount of solar energy captured during the day. Solar tracking systems are generally classified into single-axis and dual-axis tracking systems. Single-axis trackers allow the panel to rotate in one direction, typically from east to west, following the daily movement of the sun across the sky. Although single-axis trackers improve energy capture compared to fixed panels, they cannot fully compensate for changes in the sun's elevation angle. Dual-axis solar trackers, on the other hand, allow movement in both horizontal and vertical directions, enabling more precise alignment with the sun's position throughout the day and across different seasons.

This project addresses the problem by designing and implementing a Dual Axis Solar Tracker using Arduino, LDR sensors, and servo motors. The system uses multiple Light Dependent Resistors to detect variations in sunlight intensity from different directions. These sensor readings are processed by the Arduino microcontroller, which determines the direction of maximum light intensity and sends control signals to servo motors that rotate the solar panel. One servo motor controls the horizontal movement of the panel, while the other controls the vertical tilt, allowing the system to track the sun in two axes.

1.3 Motivation:

The motivation behind this project arises from the increasing demand for clean, renewable, and sustainable sources of energy. With the rapid growth of population and industrialization, the global energy demand has been rising continuously. Traditional energy sources such as coal, petroleum, and natural gas are limited in supply and contribute significantly to environmental pollution and climate change. These fossil fuels release harmful greenhouse gases into the atmosphere, which lead to global warming and environmental degradation. As a result, there is a strong need to explore alternative energy sources that are environmentally friendly, sustainable, and capable of meeting future energy requirements. Among the various renewable energy sources available today, solar energy stands out as one of the most promising and widely accessible resources because it is abundant, freely available, and clean.

Solar energy systems convert sunlight into electrical energy using photovoltaic panels. These systems are increasingly being adopted in residential, commercial, and industrial sectors due to their ability to generate electricity without producing harmful emissions. However, despite the advantages of solar energy, the efficiency of solar panels remains a major concern. In most conventional installations, solar panels are mounted in a fixed position and cannot follow the movement of the sun throughout the day. Because the sun changes its position from east to west and its elevation angle varies continuously, fixed solar panels are unable to capture the maximum possible solar radiation. This results in reduced energy generation and inefficient utilization of available sunlight.

Ultimately, the motivation behind this work is to demonstrate that AI-driven solutions, when built on robust datasets and carefully selected models, can significantly improve the accuracy, speed, and scalability of critical agricultural tasks like species classification. This has the potential to not only reduce labor and operational costs but also to foster innovation in agro-industrial processes that are traditionally resistant to change.

1.4 Objective:

The primary objective of this project is to design and implement a dual axis solar tracking system that can automatically adjust the orientation of a solar panel to maximize the amount of sunlight received throughout the day. The system aims to improve the efficiency of solar energy conversion by ensuring that the solar panel continuously faces the direction of maximum solar radiation. In order to achieve this goal, the project focuses on integrating light sensors, a microcontroller, and motor control mechanisms to create an automated solar tracking system.

Another important objective of this project is to develop a cost-effective and easy-to-implement solar tracking mechanism using commonly available electronic components. The system uses Light Dependent Resistor sensors to detect variations in sunlight intensity from different directions. These sensors provide input signals to the Arduino microcontroller, which processes the sensor data and determines the direction where the sunlight intensity is highest. Based on this information, the microcontroller generates control signals that drive servo motors to adjust the position of the solar panel. One servo motor is responsible for controlling the horizontal movement of the panel, while the second servo motor controls the vertical tilt. This dual-axis mechanism enables the solar panel to track the sun both horizontally and vertically, allowing for more accurate alignment with the sun's position.

The project also aims to demonstrate the practical use of embedded systems and microcontroller programming in renewable energy applications. By using the Arduino platform, the system can be easily programmed and modified to improve performance or incorporate additional features. Another objective is to create a system that can operate automatically without human intervention, thereby reducing manual adjustments and ensuring consistent tracking performance. Through the implementation of this project, the overall goal is to enhance the efficiency of solar power generation while promoting the use of renewable energy technologies in an accessible and affordable manner.

These sensors provide input signals to the Arduino microcontroller, which processes the sensor data and determines the direction where the sunlight intensity is highest. Based on this information, the microcontroller generates control signals that drive servo motors to adjust the position of the solar panel. One servo motor is responsible for controlling the horizontal movement of the panel, while the second servo motor controls the vertical tilt.

By continuously aligning solar panels with the direction of maximum sunlight, solar tracking systems can significantly increase the amount of electrical energy generated compared to fixed solar panel installations. This improvement in efficiency can lead to better utilization of available solar resources and reduced dependence on conventional energy sources.

The use of automated tracking systems allows solar panels to capture more sunlight throughout the day, resulting in higher energy production and improved return on investment. Additionally, solar tracking systems can be used in solar-powered street lighting, where efficient energy generation is necessary to ensure reliable operation during nighttime hours.

1.5 Applications:

The dual axis solar tracking system developed in this project has numerous practical applications in the field of renewable energy and automated energy systems. As the global demand for sustainable and clean energy sources continues to grow, improving the efficiency of solar power generation has become an important area of research and development. Solar tracking systems play a significant role in maximizing the amount of solar energy captured by photovoltaic panels. By continuously adjusting the orientation of the solar panel toward the direction of maximum sunlight, dual axis solar trackers significantly enhance the performance of solar energy systems. The use of sensors, microcontrollers, and servo motors in such systems allows solar panels to automatically follow the movement of the sun throughout the day, thereby increasing energy output compared to fixed solar installations. The implementation of this technology can be applied in several areas where efficient solar energy utilization is required.

Another important application of dual axis solar tracking systems is in residential and small-scale solar energy installations. Homeowners and small businesses that use solar panels for electricity generation can benefit from solar tracking technology by increasing the performance of their solar systems. The use of automated tracking systems allows solar panels to capture more sunlight throughout the day, resulting in higher energy production and improved return on investment. Additionally, solar tracking systems can be used in solar-powered street lighting, where efficient energy generation is necessary to ensure reliable operation during nighttimehours.

Residential Solar Energy Systems:

Dual axis solar tracking systems are widely used in solar power generation plants to increase the efficiency of photovoltaic panels. In large solar farms, maximizing energy generation is essential to ensure that the power plant produces the highest possible electricity output. Fixed solar panels only receive optimal sunlight for a limited period during the day, whereas solar trackers continuously adjust the panel orientation to maintain direct exposure to sunlight. By following the movement of the sun from east to west and adjusting the tilt angle vertically, dual axis trackers can increase energy generation by approximately thirty to forty percent compared to stationary solar panels. This improvement in efficiency leads to better utilization of available solar radiation and increases the overall productivity of solar power plants. As renewable energy becomes a key component of global energy strategies, the use of advanced solar tracking technologies is expected to grow in both small-scale and large-scale power generation systems.

Solar tracking systems can also be applied in solar-powered street lighting systems to improve their efficiency and reliability. Solar street lights are commonly used in urban and rural areas to provide lighting during nighttime hours using energy stored in batteries charged by solar panels during the day. In many cases, the solar panels used in these systems are fixed in position, which limits the amount of solar energy that can be collected during daylight hours. By using a dual axis solar tracker, the solar panel can follow the sun’s movement and capture more sunlight throughout the day. This increased energy collection ensures that the battery receives sufficient charge to power the street lights during nighttime operation. As a result, solar-powered lighting systems become more reliable and capable of operating efficiently even in locations with limited sunlight exposure.

Agricultural Irrigation Systems:

Dual axis solar tracking technology can also be applied in agricultural irrigation systems that rely on solar power to operate water pumps. In many rural and remote areas, farmers depend on solar energy to run irrigation equipment because access to conventional electricity may be limited or unavailable. However, the efficiency of solar-powered irrigation systems depends heavily on the amount of energy generated by solar panels. By incorporating a solar tracking mechanism, the solar panel can receive maximum sunlight throughout the day, allowing the system to generate more electrical power for operating irrigation pumps. This improved energy generation ensures that sufficient water can be pumped for agricultural purposes, supporting crop growth and improving farming productivity. Solar tracking systems therefore play an important role in promoting sustainable agricultural practices and reducing dependence on fossil fuel-based energy sources.

Educational and Research Applications:

Dual axis solar tracking systems are also widely used in educational institutions and research laboratories to demonstrate the principles of renewable energy, embedded systems, and automation technology. Engineering students and researchers can use solar tracking projects to understand how sensors, microcontrollers, and motors work together to create intelligent automated systems. Such projects provide hands-on experience in programming microcontrollers, designing electronic circuits, and integrating hardware and software components. By studying solar tracking systems, students gain valuable knowledge about energy management, renewable energy technologies, and control systems. In addition, research institutions may use solar tracking systems to conduct experiments on improving photovoltaic efficiency, developing advanced tracking algorithms, and designing innovative renewable energy solutions.

Remote Power Supply Systems:

Solar tracking technology is particularly useful in remote locations where access to conventional electricity infrastructure is limited. Areas such as rural villages, remote monitoring stations, communication towers, and weather observation systems often rely on solar energy as their primary power source. In such locations, maximizing solar energy generation is critical to ensure reliable operation of electronic equipment and communication systems. A dual axis solar tracker can significantly increase the amount of solar energy captured by photovoltaic panels, ensuring that sufficient electrical power is available for operating devices in remote areas. This application is especially important for environmental monitoring systems, satellite communication equipment, and remote research facilities where consistent power supply is essential.

Smart Renewable Energy Systems:

With the advancement of smart technologies and automation, solar tracking systems are becoming an integral part of intelligent renewable energy systems. By combining solar trackers with sensors, microcontrollers, and data monitoring systems, it is possible to develop smart solar energy solutions that optimize power generation based on environmental conditions. These systems can monitor sunlight intensity, weather patterns, and panel performance to automatically adjust the orientation of solar panels for maximum efficiency. The integration of solar tracking technology with Internet of Things (IoT) platforms and energy management systems can further enhance the performance of renewable energy installations. Such smart renewable energy systems contribute to efficient power generation, improved energy management, and sustainable development.

CHAPTER-2

LITERATURE SURVEY

The rapid growth in global energy demand and the increasing concerns regarding environmental sustainability have led researchers to explore renewable energy technologies as alternatives to conventional fossil fuels. Among the various renewable energy sources available, solar energy has emerged as one of the most promising and widely used resources due to its abundance, reliability, and environmentally friendly nature. Solar energy systems convert sunlight into electrical energy through photovoltaic cells, which form the core of solar panels. However, the efficiency of solar panels depends largely on the amount of solar radiation they receive. In conventional solar energy systems, solar panels are installed in a fixed position, which limits their ability to capture maximum sunlight throughout the day. As the sun moves from east to west across the sky, the angle of sunlight incident on the panel changes continuously, resulting in reduced energy generation. To address this issue, researchers have developed solar tracking systems that automatically adjust the orientation of solar panels in order to maintain optimal alignment with the sun.Early studies in solar tracking technology focused primarily on single-axis tracking systems. These systems allow solar panels to rotate along one axis, typically following the daily east-to-west movement of the sun. Researchers found that single-axis solar trackers can significantly improve the efficiency of solar panels compared to fixed installations. By maintaining a better alignment with the sun during daylight hours, single-axis tracking systems can increase energy output by approximately twenty to twenty-five percent. Although single-axis tracking provides a notable improvement in performance, it does not fully compensate for variations in the sun's elevation angle throughout the year. Seasonal changes cause the sun to appear higher or lower in the sky, which means that a solar panel must also adjust its tilt angle to maintain optimal exposure to sunlight. As a result, researchers began to explore dual-axis solar tracking systems that allow movement in both horizontal and vertical directions. LDR’s are the most commonly used sensors in solar tracking systemsbecause they are inexpensive, easy to integrate with electronic circuits, and highly sensitive to changes in light intensity. LDR sensors change there electrical resistance.

Dual-axis solar trackers provide a more accurate method for maintaining solar panel alignment with the sun. These systems use two independent axes of movement to track the sun's position across the sky and adjust the orientation of the solar panel accordingly. Several studies have demonstrated that dual-axis solar trackers can increase energy generation by approximately thirty to forty percent compared to stationary panels. This significant improvement in efficiency has made dual-axis tracking systems an attractive solution for both small-scale and large-scale solar energy applications. Researchers have proposed various mechanical designs and control mechanisms to implement dual-axis tracking systems. Some systems use complex mechanical structures and high-power motors to rotate large solar panels, while others focus on simpler designs that can be implemented using low-cost electronic components. One of the key aspects of solar tracking systems is the method used to detect the direction of sunlight. In early tracking systems, solar panel movement was controlled using pre-programmed timers that moved the panel at fixed intervals throughout the day. These timer-based systems were relatively simple to implement but lacked accuracy because they did not respond to real-time variations in sunlight intensity caused by weather conditions such as clouds or atmospheric changes. To overcome this limitation, researchers introduced sensor-based tracking systems that use light sensors to detect the intensity of sunlight from different directions.

CHAPTER 3 EXIST-
ING SYSTEM

3.1 Traditional methodology

In traditional solar energy systems, photovoltaic panels are installed in a fixed position where the tilt angle and orientation are predetermined during installation. This orientation is usually selected based on geographical location, seasonal sunlight patterns, and the average direction of maximum solar radiation in a specific region. The solar panel is generally tilted toward the equator to capture maximum sunlight during midday hours when solar radiation is strongest. However, once the solar panel is installed, its position remains unchanged throughout the day and across different seasons.

In this traditional approach, solar panels rely solely on the natural movement of the sun to provide varying levels of sunlight exposure throughout the day. As the sun rises in the east, the sunlight strikes the solar panel at an inclined angle, which means that the intensity of solar radiation received by the panel is relatively low. During midday, when the sun is positioned almost directly above the panel, the angle of incidence becomes closer to perpendicular, resulting in higher energy generation. Later in the afternoon and evening, as the sun moves toward the west, the angle between the solar panel and the incoming sunlight increases again, causing a reduction in energy absorption.

Because photovoltaic cells produce maximum electrical power when sunlight falls directly perpendicular to their surface, any deviation from this optimal angle reduces the amount of solar radiation absorbed by the panel. In a fixed solar panel system, the panel can only maintain optimal alignment with the sun for a limited period of time during the day. For the remaining hours, the sunlight falls at an angle that decreases the efficiency of energy conversion. As a result, a significant portion of available solar energy remains unused.

Traditional solar power systems also rely heavily on manual design considerations during installation. Engineers determine the optimal tilt angle and orientation based on long-term solar radiation data for the specific location. While this approach helps achieve a reasonable level of efficiency, it cannot account for real-time variations in sunlight direction caused by environmental factors such as cloud cover, seasonal changes, and atmospheric conditions. Therefore, traditional fixed solar panels are unable to adapt dynamically to changes in sunlight conditions.

Overall, the traditional methodology of solar energy systems involves installing fixed solar panels that remain stationary and depend on the natural movement of the sun to generate ele ctricity. Although this method is simple and widely used, it does not utilize the full potential of available solar radiation.

Limitations of the Traditional System

Although traditional fixed solar panel systems are simple and widely used, they have several limitations that reduce their efficiency and overall performance.

One of the major limitations of the traditional system is its inability to track the movement of the sun. Since the solar panel remains fixed in position, it cannot adjust its orientation to follow the sun as it moves across the sky. This results in suboptimal sunlight exposure during most hours of the day, reducing the total energy generated by the panel.

Another important limitation is the reduced efficiency of solar energy conversion. Photovoltaic cells generate maximum electrical power when sunlight strikes the panel surface at a perpendicular angle. In fixed systems, the angle of sunlight continuously changes as the sun moves from east to west. Because the panel cannot adjust its position, the sunlight often falls at an oblique angle, leading to a significant loss of potential energy.

Seasonal variations in the sun’s position also affect the performance of traditional solar systems. During different times of the year, the elevation angle of the sun changes due to the tilt of the Earth's axis. Fixed solar panels cannot adapt to these seasonal changes, which further reduces their efficiency in capturing solar radiation.

Another drawback of traditional solar systems is the inefficient utilization of available sunlight during early morning and late afternoon hours. During these periods, sunlight reaches the panel at a shallow angle, resulting in lower energy generation. Since the panel orientation remains fixed, it cannot reposition itself to capture sunlight more effectively during these times.

CHAPTER 4

PROPOSED SYSTEM

Solar tracking technology is also useful in agricultural applications, particularly in solar-powered irrigation systems. Farmers who rely on solar energy to operate water pumps can benefit from improved energy generation provided by solar tracking systems. By maximizing solar panel efficiency, these systems ensure that sufficient electrical energy is available to power irrigation equipment, thereby supporting sustainable agricultural practices. Furthermore, dual axis solar trackers can be used in research laboratories, educational institutions, and engineering projects to demonstrate the principles of renewable energy, embedded systems, and automation technology. These systems serve as valuable teaching tools that help students understand the integration of sensors, microcontrollers, and mechanical components in real-world applications. The system is also cost-effective because it uses simple and inexpensive components such as Arduino microcontrollers, LDR sensors, and servo motors. These components are widely available and easy to integrate, making the system suitable for educational projects and small-scale solar installations.

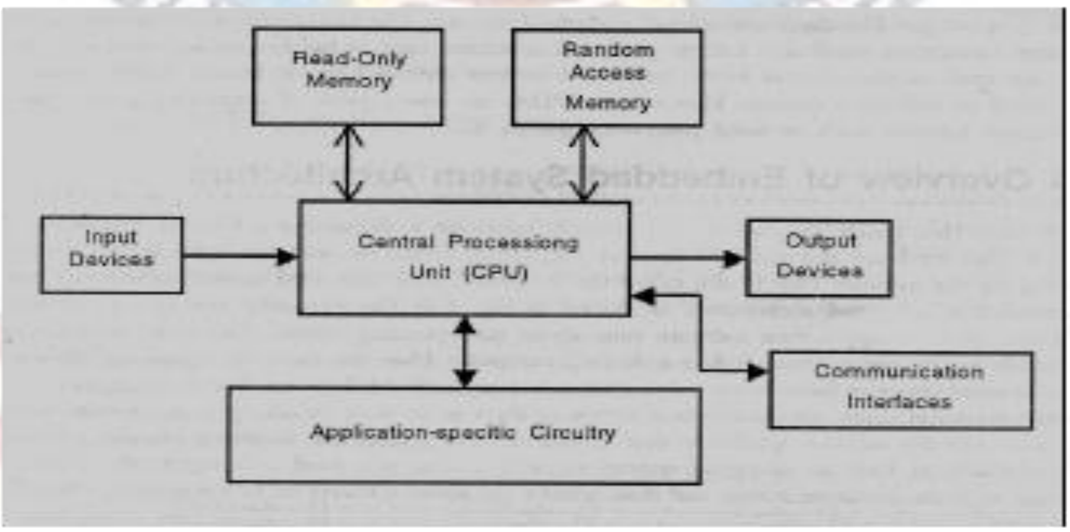

Fig 4.1: Proposed Block Diagram

4.1 Overview

Fig 4.1 Describes that

The proposed system in this project is designed to improve the efficiency of solar energy generation by automatically tracking the movement of the sun throughout the day. Traditional solar panels are fixed in position and cannot adjust their orientation according to the direction of sunlight. As a result, they receive optimal sunlight only during a limited period of time, which reduces the overall energy conversion efficiency. In order to overcome this limitation, the proposed system introduces an automated dual axis solar tracking mechanism that continuously aligns the solar panel toward the direction of maximum sunlight.

1. CENTRAL PROCESSING UNIT:

At the core of the system is the Central Processing Unit (CPU), which performs the main cont rol and decision-making functions. In this project, the Arduino microcontroller acts as the CPU of the system. It receives input signals from the sensors, processes the data using pro grammed instructions, and generates control signals to operate the servo motors that adjust the position of the solar panel. The CPU coordinates the overall functioning of the system by managing communication between different hardware components

2. READ ONLY MEMORY (ROM):

The Read Only Memory (ROM) stores the program instructions required for the operation of the system. These instructions include the algorithm used to compare sensor values, determine the direction of maximum sunlight, and control the movement of the solar panel. Since the program stored in ROM does not change during normal operation, it ensures reliable and consistent system performance.provides insight into how the data might influence the classifier’s performance.

3. RANDOM ACCESS MEMORY (RAM)

The Random Access Memory (RAM) is used as temporary storage during system operation. It stores intermediate data such as sensor readings, calculated values, and control parameters required for the execution of the tracking algorithm. RAM allows the CPU to quickly access and modify data while processing information from the sensors.

4. INPUT DEVICES.

The Input Devices in the proposed system consist of Light Dependent Resistor (LDR) sensors that detect the intensity of sunlight from different directions. These sensors generate analog signals that vary depending on the amount of light falling on them. The signals from the sen sors are sent to the microcontroller, where they are converted into digital values and analyzed to determine the direction of maximum sunlight.

5. APPLICATION-SPECIFIC CIRCUITRY.

The **Application-Specific Circuitry** includes the electronic circuits used to interface sensors and actuators with the microcontroller. This circuitry may include voltage divider networks for LDR sensors, motor driver circuits, and signal conditioning components that ensure proper operation of the system. These circuits play an important role in enabling communication between hardware components and the control unit.

6. OUTPUT DEVICES.

The **Output Devices** in the system include servo motors that control the movement of the solar panel. Based on the processed sensor data, the microcontroller sends control signals to the servo motors to rotate the solar panel horizontally and vertically. This dual-axis movement allows the solar panel to continuously track the position of the sun and maintain optimal alignment for maximum energy absorption.

4.2 CENTRAL PROCESSING UNIT.

The **central processing unit (CPU)** acts as the brain of the system and is responsible for controlling the overall operation of the solar tracker. In this project, the Arduino microcontroller functions as the CPU. It receives the analog input signals from the LDR sensors, converts them into digital values, and processes the data using the programmed algorithm. The microcontroller compares the sensor readings and determines whether the solar panel needs to move left, right, upward, or downward in order to align with the direction of maximum sunlight. After performing these calculations, the CPU sends control signals to the output devices that move the solar panel accordingly.

The sensor data is sent to the microcontroller, where it is processed and compared. If the microcontroller detects a difference in light intensity between sensors, it determines the direction in which the solar panel needs to move. Control signals are then sent to the servo motors to adjust the panel orientation accordingly.

4.3 INPUT DEVICES.

The input devices play an important role in detecting environmental conditions and providing data to the control system. In the proposed solar tracking system, Light Dependent Resistor (LDR) sensors act as the primary input devices. These sensors are sensitive to light and their resistance changes according to the intensity of light falling on them. Four LDR sensors are typically placed around the solar panel in different directions. When sunlight falls unevenly on these sensors, they generate different voltage signals which are sent to the microcontroller. The microcontroller continuously reads these signals to determine which direction has the highest light intensity. This information is used to adjust the orientation of the solar pan el. These sensors are sensitive to light and their resistance changes according to the intensity of light falling on them. Four LDR sensors are typically placed around the solar panel in dif ferent directions. When sunlight falls unevenly on these sensors, they generate different volt age signals which are sent to the microcontroller. The microcontroller continuously reads these signals to determine which direction has the highest light intensity. This information is used to adjust the orientation of the solar panel. The sensor data is sent to the microcontroller, where it is processed and compared. If the microcontroller detects a difference in light intensi ty between sensors, it determines the direction in which the solar panel needs to move. Con trol signals are then sent to the servo motors to adjust the panel orientation accordingly.
In order to overcome this limitation, the proposed system introduces an automated dual axis solar tracking mechanism that continuously aligns the solar panel toward the direction of maximum sunlight.

4.4 COMMUNICATION INTERFACE

The **communication interfaces** allow the microcontroller to communicate with external devices for monitoring, debugging, or system expansion. In embedded systems such as Arduino-based projects, communication interfaces may include serial communication through USB ports or wireless communication modules. These interfaces enable data transfer between the microcontroller and a computer or monitoring system. Although communication interfaces may not be actively used in basic solar tracking systems, they provide the flexibility to integrate additional features such as remote monitoring or data logging in future improvements.

The overall working of the block diagram begins when sunlight falls on the LDR sensors. These sensors detect the intensity of light from different directions and send corresponding signals to the microcontroller. The microcontroller processes these signals and determines the direction in which the solar panel should move. Based on the processed data, control signals are sent to the servo motors to adjust the panel orientation. This process continues continuously throughout the day, allowing the solar panel to track the movement of the sun and maintain optimal alignment for maximum energy absorption.

In this project, the application-specific circuitry includes voltage divider circuits used with LDR sensors, power supply circuits, and motor control interfaces that connect the microcontroller to the servo motors. The LDR sensors are connected in voltage divider configurations with resistors to convert variations in light intensity into measurable voltage signals. These signals are then sent to the analog input pins of the Arduino microcontroller for processing.

In this project, the application-specific circuitry includes voltage divider circuits used with LDR sensors, power supply circuits, and motor control interfaces that connect the microcontroller to the servo motors. The LDR sensors are connected in voltage divider configurations with resistors to convert variations in light intensity into measurable voltage signals. These signals are then sent to the analog input pins of the Arduino microcontroller for processing.

RAM allows the CPU to quickly access and modify data during program execution. This temporary storage is essential for performing real-time calculations and ensuring smooth operation of the solar tracking system.

model very well and its training accuracy is also very high, but we provide a new dataset to it, then it will decrease the performance. So we always try to make a machine learning model which performs well with the training set and also with the test dataset. Here, we can define these datasets as:

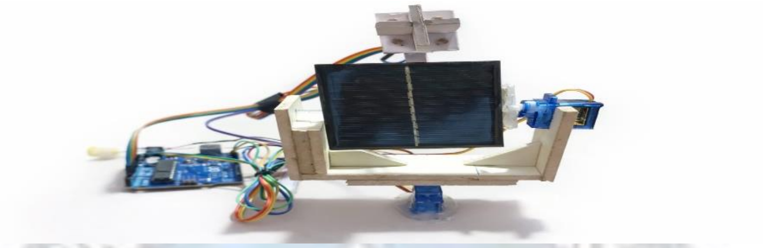

Fig:4.2: Dual Axis solar Aurdino Project.

Fig 4.2 describes that

4.5 Working Of This Model.

The working principle of the dual axis solar tracking system is based on the continuous moni toring of sunlight intensity and the automatic adjustment of the solar panel orientation to maintain maximum exposure to solar radiation. The system uses light sensors, a microcontrol ler, and servo motors to track the movement of the sun and ensure that the solar panel remains aligned with the direction of maximum sunlight throughout the day. By maintaining the solar panel perpendicular to the sun's rays, the system maximizes the amount of solar energy ab sorbed by the photovoltaic cells, thereby improving the efficiency of power generation.

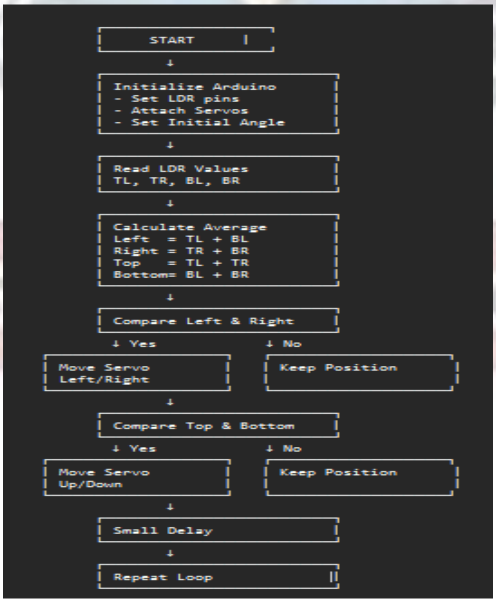

Fig:4.3: Workflow of Existing Model

The flowchart represents the logical sequence of operations performed by the solar tracking system. It provides a clear representation of how the system processes sensor data and controls the movement of the solar panel. The flowchart helps in understanding the algorithm used in the Arduino program for automatic solar tracking.

This matrix contains the extracted features from the images, such as pixel intensities, texture, shape, and color histograms. These are the key attributes learned by the classifier to differentiate between Kirmizi and Siirt pistachios.The operation of the system begins with the initialization stage, where the microcontroller initializes the required components such as sensor inputs and servo motor outputs.

During this stage, the Arduino sets the initial position of the servo motors, usually placing the solar panel in a neutral position facing upward toward the sun. The system then enters the main operating loop where it continuously monitors the sensor readings. In the next step, the system reads the values from the LDR sensors.

The four LDR sensors are connected to the analog input pins of the Arduino microcontroller. Each sensor produces a voltage signal based on the intensity of sunlight falling on it. The Arduino reads these analog signals and converts them into digital values that can be used for further processing. In the next step, the system reads the values from the LDR sensors.

If the difference in light intensity between the left and right sensors exceeds a predefined threshold value, the microcontroller decides that the solar panel needs to move horizontally toward the direction receiving higher light intensity. Similarly, if the difference between the top and bottom sensors exceeds the threshold value, the panel must tilt vertically toward the direction of stronger sunlight.

Once the direction of movement is determined, the Arduino sends control signals to the servo motors. The servo motors rotate the solar panel by small increments until the sensor readings become balanced. When the sensor readings are nearly equal, it indicates that the solar panel is aligned with the direction of maximum sunlight.

After adjusting the position of the solar panel, the system waits for a short delay before repeating the process. This delay prevents unnecessary rapid movements of the motors and allows the system to stabilize before taking the next reading. The system then returns to the beginning of the loop and repeats the process of reading sensor values, comparing light intensities, and adjusting the panel orientation.

This continuous cycle ensures that the solar panel tracks the movement of the sun throughout the day. The system then returns to the beginning of the loop and repeats the process of reading sensor values, comparing light intensities, and adjusting the panel orientation. This continuous cycle ensures that the solar panel tracks the movement of the sun throughout the day. The power supply unit provides the electrical energy required to operate the entire system. The Arduino microcontroller can be powered using a USB connection or a regulated external power supply. However, servo motors typically require a stable 5V power source to operate efficiently. Therefore, an external power supply is often used to power the servo motors to ensure reliable performance. Proper power management is important in solar tracking systems because unstable voltage levels can affect the performance of the sensors and actuators. The solar panel is the main component responsible for converting solar energy into electrical energy. Photovoltaic cells within the solar panel absorb sunlight and generate electrical current through the photovoltaic effect.

In the proposed system, the solar panel is mounted on a mechanical structure that allows it to rotate and tilt based on the movement of the servo motors. As the solar tracker continuously adjusts the panel orientation to follow the sun, the solar panel receives maximum solar radiation, resulting in improved energy generation compared to fixed solar panel systems. The solar panel is the main component responsible for converting solar energy into electrical energy. Photovoltaic cells within the solar panel absorb sunlight and generate electrical current through the photovoltaic effect. In the proposed system, the solar panel is mounted on a mechanical structure that allows it to rotate and tilt based on the movement of the servo motors. As the solar tracker continuously adjusts the panel orientation to follow the sun, the solar panel receives maximum solar radiation, resulting in improved energy generation compared to fixed solar panel systems.

4.5.1 HARWARE COMPONENTS DISCRIPTION

The proposed dual axis solar tracking system consists of several hardware components that work together to detect sunlight intensity, process sensor data, and control the movement of the solar panel. These components include the Arduino Uno microcontroller, Light Dependent Resistor (LDR) sensors, servo motors, resistors, power supply unit, and the solar panel. Each component plays an important role in ensuring the proper functioning of the solar tracking system.

4.5.2 Arduino Uno Microcontroller.

The Arduino Uno microcontroller serves as the main control unit of the system. It is responsible for reading the input signals from the sensors, processing the data using the programmed algorithm, and generating control signals for the servo motors. The Arduino Uno is based on the ATmega328P microcontroller and contains both digital and analog input/output pins that allow it to interface with various electronic components. In the proposed system, the analog input pins of the Arduino are used to read the voltage signals generated by the LDR sensors, while the digital PWM pins are used to control the servo motors. The Arduino microcontroller operates according to the program written in the Arduino Integrated Development Environment (Arduino IDE), which allows the system to perform automatic solar tracking.

4.5.3 Light Dependent Resistor.

The **Light Dependent Resistor (LDR) sensors** are used to detect the intensity of sunlight from different directions. LDR sensors are photoresistors whose electrical resistance changes depending on the amount of light falling on their surface. When exposed to bright light, the resistance of the LDR decreases, and when the light intensity decreases, the resistance increases. In the proposed system, four LDR sensors are placed at different positions around the solar panel. These sensors detect variations in sunlight intensity and provide analog signals to the Arduino microcontroller. By comparing the values obtained from different sensors, the microcontroller determines the direction of maximum sunlight and adjusts the position of the solar panel accordingly. The signals generated by the LDR sensors are analog signals that vary according to the intensity of light. However, these signals cannot be directly interpreted by the microcontroller without proper signal conditioning. Therefore, the next stage in the block diagram involves converting these resistance changes into measurable voltage signals.

4.5.4 Servo Motors.

The **servo motors** are used as actuators to move the solar panel in both horizontal and vertical directions. Servo motors are preferred for this application because they provide precise control over angular position. Unlike conventional DC motors, servo motors can rotate to a specific angle based on the control signal received from the microcontroller. In the dual axis solar tracking system, two servo motors are used. One servo motor controls the horizontal rotation of the solar panel, allowing it to move from east to west following the movement of the sun. The second servo motor controls the vertical tilt of the panel, enabling it to adjust its angle according to the elevation of the sun in the sky. This dual-axis movement ensures that the solar panel remains aligned with the direction of maximum sunlight.

4.5.5 Resistor.

Fig 4.4 Description :

4.5.6: Sunlight As The Primary Energy Source

Sunlight is the primary energy source for the solar tracking system. Solar energy is one of the most abundant and renewable sources of energy available on Earth. The energy from sunlight is converted into electrical energy using photovoltaic cells present in solar panels. However, the efficiency of solar panels depends heavily on the angle at which sunlight strikes the panel surface. Maximum electrical power is generated when sunlight falls perpendicular to the surface of the solar panel. As the sun moves across the sky from east to west during the day, the angle of sunlight changes continuously. If the solar panel remains fixed, it cannot maintain this optimal alignment with the sun, resulting in reduced energy conversion efficiency.

4.5.7: LDR Sensor Of Sunlight detection

The first major component in the block diagram is the LDR sensor module, which consists of four Light Dependent Resistors labeled as TL (Top Left), TR (Top Right), BL (Bottom Left), and BR (Bottom Right). These sensors are placed at four different positions around the solar panel so that they can detect sunlight from different directions. LDR sensors are photoresistors whose electrical resistance varies depending on the intensity of light falling on them. When the light intensity increases, the resistance of the LDR decreases, and when the light intensity decreases, the resistance increases.

By placing four LDR sensors around the solar panel, the system can detect differences in light intensity across the panel. For example, if the sensors on the right side receive more sunlight than those on the left side, it indicates that the sun is positioned toward the right. Similarly, if the top sensors receive more light than the bottom sensors, the system recognizes that the sun is positioned higher in the sky. This difference in sensor readings provides the information required to determine the direction of maximum sunlight.

4.5.8: Voltage Divider Circuit

The voltage divider circuit is used to convert the resistance variations of the LDR sensors into voltage signals that can be measured by the microcontroller. In this circuit, each LDR sensor is connected in series with a fixed resistor to form a voltage divider network. The output volt age of this circuit depends on the resistance of the LDR, which changes with light intensity. The voltage divider circuit plays an important role in ensuring that the sensor signals are properly converted into electrical signals that can be processed by the control system.

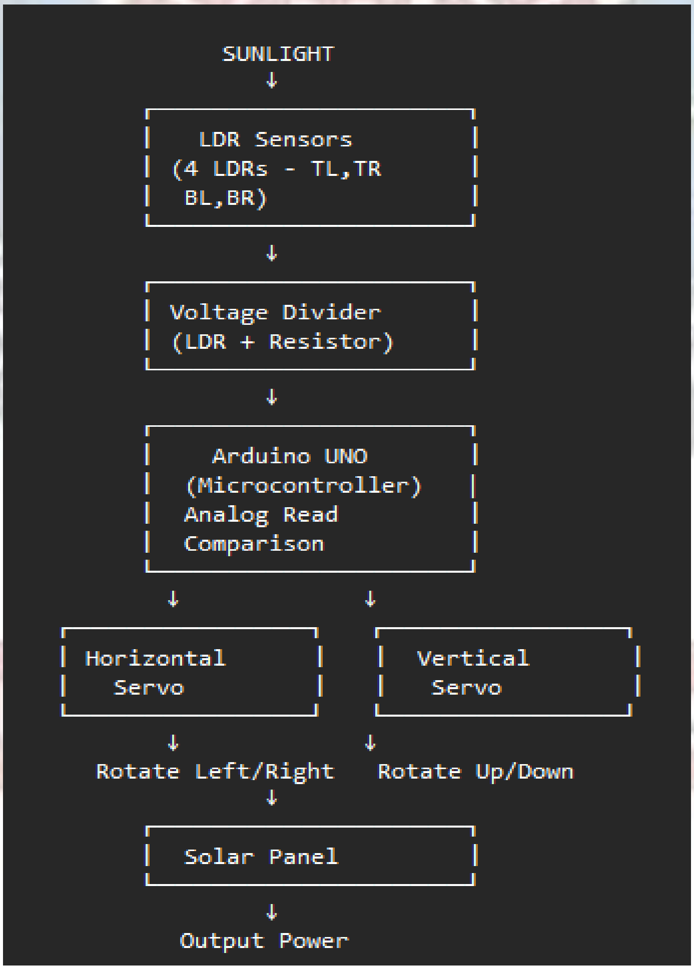

Fig 4.4 Block Diagram.

4.5.9: Arduino UNO Microcontroller

The Arduino UNO serves as the central processing unit (CPU) of the solar tracking system. It acts as the main control unit responsible for processing sensor data and controlling the movement of the solar panel. The Arduino microcontroller reads the analog voltage signals from the LDR sensors through its analog input pins. These signals are then converted into digital values using the built-in analog-to-digital converter (ADC) of the microcontroller.

After reading the sensor values, the Arduino compares the readings obtained from different LDR sensors. The microcontroller calculates the average values of the top sensors (TL and TR) and the bottom sensors (BL and BR) to determine the vertical alignment of the panel. Similarly, it com pares the average values of the left sensors (TL and BL) with the right sensors (TR and BR) to de termine the horizontal alignment.

4.5.10: Horizontal Servo Motor control.

After reading the sensor values, the Arduino compares the readings obtained from different LDR sensors. The microcontroller calculates the average values of the top sensors (TL and TR) and the bottom sensors (BL and BR) to determine the vertical alignment of the panel. Similarly, it com pares the average values of the left sensors (TL and BL) with the right sensors (TR and BR) to de termine the horizontal alignment.

This horizontal movement ensures that the solar panel remains aligned with the sun as it travels across the sky during the day.

4.5.11: Solar Panel And Power Output.

The solar panel is the final component in the block diagram. It converts solar radiation into electri cal energy using photovoltaic cells. As the dual axis tracking system continuously adjusts the orien tation of the panel to face the sun, the panel receives maximum sunlight for a longer period during the day. This increased exposure to sunlight results in higher electrical power generation compared to fixed solar panels.

The electrical energy produced by the solar panel can then be used to power electrical devices or stored in batteries for later use.By integrating sensors, microcontroller processing, and servo motor control, the dual axis solar tracking system provides an efficient and automated solution for im proving solar power generation. The block diagram therefore represents the complete functional flow of the system and demonstrates how different components work together to achieve accurate solar tracking.

CHAPTER 5
SOFTWARE REQUIREMENTS

5.1 Introduction

Software plays a crucial role in the functioning of the dual axis solar tracking system. The hardware components such as LDR sensors, servo motors, and the Arduino microcontroller require appropriate programming instructions in order to operate correctly. The software used for programming the Arduino microcontroller in this project is the Arduino Integrated Development Environment (Arduino IDE). The Arduino IDE allows users to write, compile, and upload programs to the Arduino board. These programs control the behavior of the microcontroller and enable it to read sensor inputs, process data, and generate output signals to control the servo motors.

The Arduino IDE is an open-source software platform that supports programming in a simplified version of the C and C++ programming languages. It provides a user-friendly interface that allows developers to easily write and debug programs for Arduino boards. In the proposed solar tracking system, the Arduino program continuously reads the sensor values from the LDR sensors, compares the light intensity detected by each sensor, and controls the servo motors to adjust the position of the solar panel.

The Arduino IDE is an open-source software platform that supports programming in a simplified version of the C and C++ programming languages. It provides a user-friendly interface that allows developers to easily write and debug programs for Arduino boards. In the proposed solar tracking system, the Arduino program continuously reads the sensor values from the LDR sensors, compares the light intensity detected by each sensor, and controls the servo motors to adjust the position of the solar panel. In the proposed system, the Arduino program continuously monitors the light intensity detected by four LDR sensors placed around the solar panel. The program compares the sensor values to determine which direction receives the highest amount of sunlight. Based on this comparison, the program generates control signals that drive the servo motors responsible for rotating and tilting the solar panel. By repeating this process continuously, the system ensures that the solar panel always faces the direction of maximum sunlight.

Arduino IDE

The code editor is used to write the program that defines the behavior of the microcontroller. Once the program is written, the compiler converts the code into machine language that can be executed by the microcontroller. The compiled program is then uploaded to the Arduino board using a USB cable. The IDE also includes a serial monitor, which allows users to view real-time data such as sensor readings and system status. This feature is particularly useful during system testing and debugging.

Another important feature of the Arduino IDE is its library support. Libraries are pre-written collections of code that simplify the process of interfacing with hardware components such as sensors, motors, and communication modules. By using libraries, developers can easily control complex devices without writing detailed low-level code.

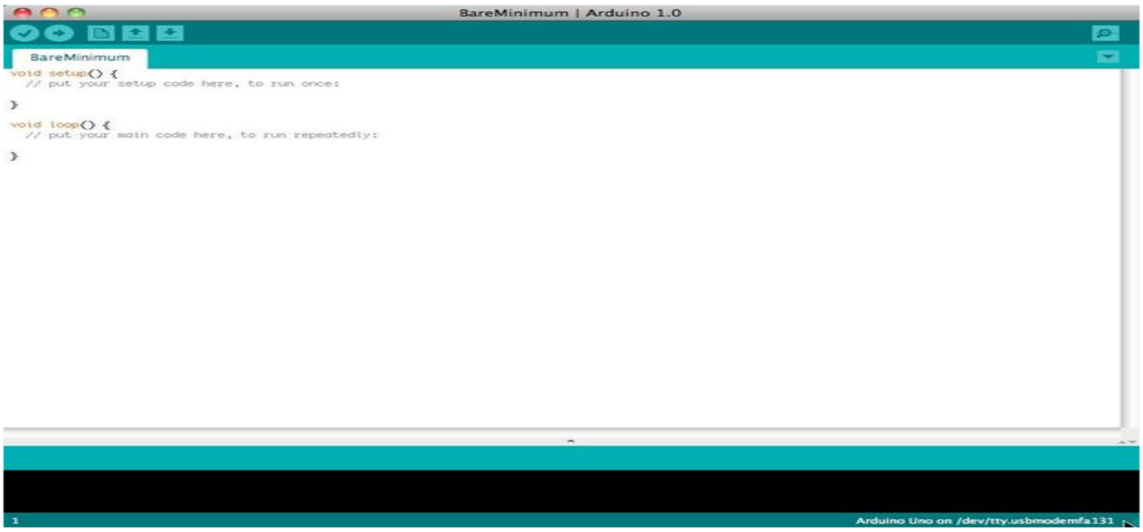

Fig 5.1 Arduino Setup

Arduino Overview

Servo Library: The most important library used in this project is the Servo library. This library allows the Arduino microcontroller to control servo motors easily by generating Pulse Width Modulation (PWM) signals. The Servo library provides functions such as attach(), write(), and detach() that allow the programmer to specify the angle of rotation for the servo motor. In the solar tracking system, two servo motors are used. One servo motor controls the horizontal movement of the solar panel, while the other controls the vertical tilt. By using the Servo library, the Arduino program can rotate the motors to specific angles based on the sensor readings.

Wire Library: The Wire library is used for communication between Arduino and other devices through the I2C communication protocol. Although it may not be required for basic solar tracking systems, it can be used in advanced implementations where additional sensors or display modules are added.

SPI Library : The SPI (Serial Peripheral Interface) library is used for high-speed communication between the microcontroller and peripheral devices. This library can be useful if the solar tracking system is expanded to include additional modules such as data storage or communication devices.

EEPROM Library: The EEPROM library allows data to be stored permanently in the Arduino memory even after power is turned off. This feature can be useful for saving system parameters such as calibration values or servo motor positions tasks. While newer versions does not add features like enhanced type checking, 0.25.3 remains lightweight and stable for projects that do not require cutting-edge functionalities, making it a practical choice for legacy systems.

SoftwareSerial Library: The SoftwareSerial library enables serial communication on digital pins other than the default serial pins. This can be useful if additional communication modules such as Bluetooth or GSM modules are integrated with the solar tracking system. The SoftwareSerial library enables serial communication on digital pins other than the default serial pins. This can be useful if additional communication modules such as Bluetooth or GSM modules are integrated with the solar tracking system.

Microcontroller	:	ATmega328
Operating Voltage	:	5V
Input Voltage (recommended)	:	7-12V
Input Voltage (limits)	:	6-20V
Digital I/O Pins	:	14 (of which 6 provide PWM output)
Analog Input Pins	:	6
DC Current per I/O Pin	:	40 mA
DC Current for 3.3V Pin	:	50 mA
Flash Memory used by bootloader	:	32 KB (ATmega328) of which 0.5 KB
SRAM	:	2 KB (ATmega328)
EEPROM	:	1 KB (ATmega328)
Clock Speed	:	16 MHz
Length	:	68.6 mm
Width	:	53.4 mm

USB Plug & External Power Supply Plug : Every Arduino board needs a way to be connected to a power source. The Arduino Uno can be powered from a USB cable coming from your computer or a wall power supply that is terminated in a barrel jack. The power source is selected automatically. The USB connection is also how you will load code onto your Arduino board. Please on my other post on how to program with Arduino can be found in Installing and Programming Arduino.

NOTE: The board can operate on an external supply of **6 to 20 volts**. If supplied with less than **7V**, however, the **5V** pin may supply less than five volts and the board may be unstable. If using more than **12V**, the voltage regulator may overheat and damage the board. The recommended range is **7 to 12 Volts**.

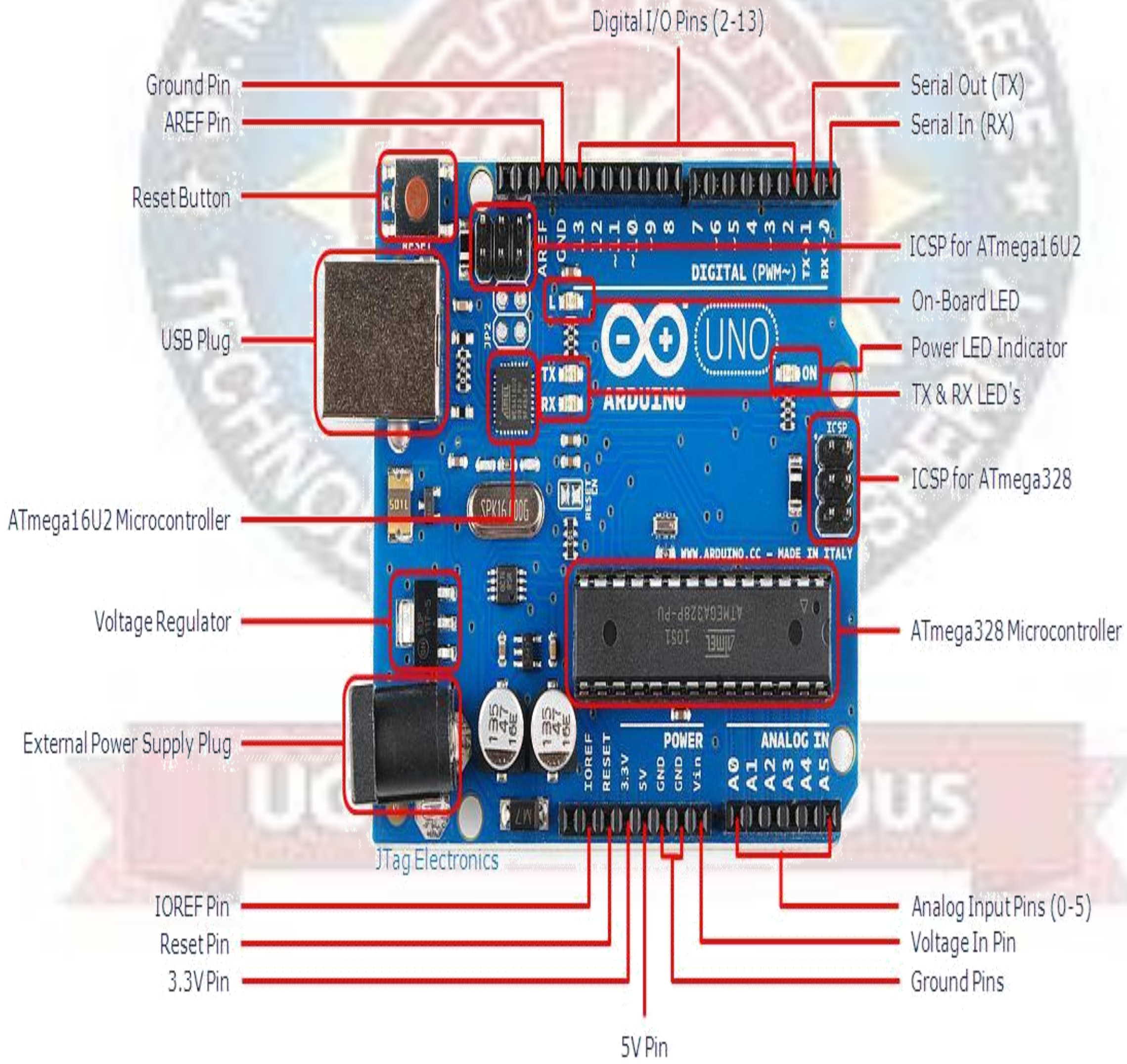

Fig 5.2 Arudino UNO Board

5.2 Programming Structure

The **setup() function** runs only once when the Arduino board is powered on or reset. In this function, the system initializes the required components and prepares them for operation. For example, the servo motors are attached to their respective pins and the initial position of the solar panel is set.The loop() function runs continuously after the setup function is completed. This function contains the main logic of the solar tracking system. Inside the loop function, the Arduino reads the sensor values, compares the light intensity detected by each sensor, and adjusts the position of the solar panel accordingly. Because the loop function repeats continuously, the solar panel is able to track the movement of the sun in real time.

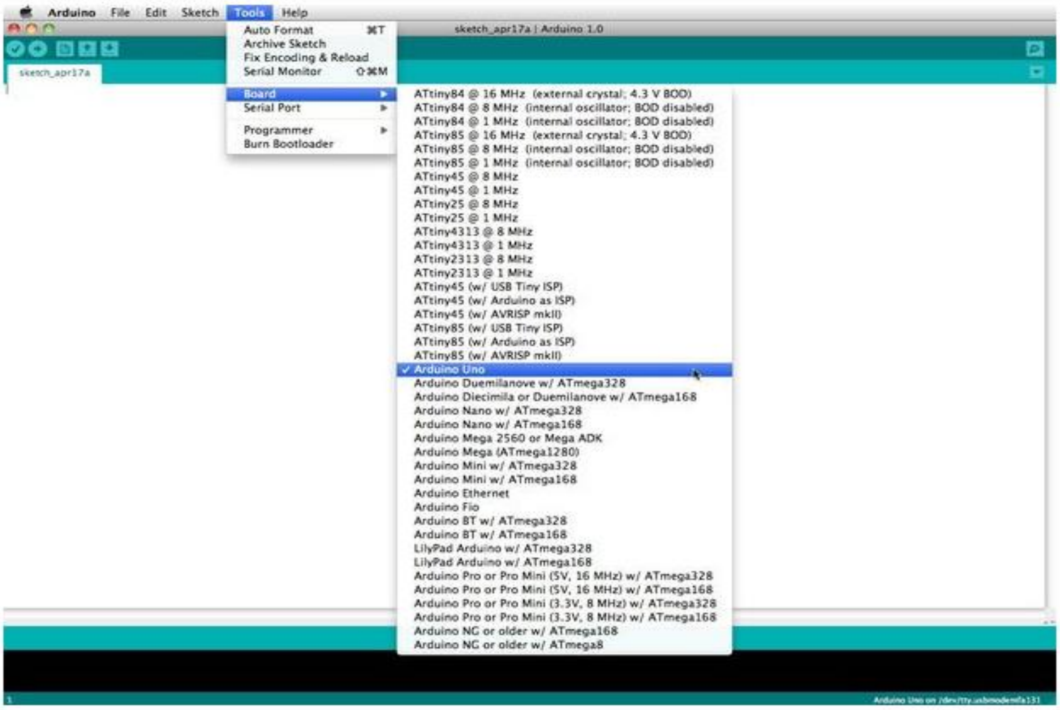

Fig 5.3 Arudino Selection

5.3 Sensor Data Acquisition

The LDR sensors used in the solar tracking system produce analog voltage signals that vary depending on the intensity of light. These signals are connected to the analog input pins of the Arduino microcontroller. The Arduino reads these signals using the analogRead() function. The analog values obtained from the sensors typically range from 0 to 1023. These values represent the intensity of light detected by each sensor.

The Arduino program processes these values to determine which sensor receives the highest amount of sunlight. In the proposed system, the four LDR sensors are grouped into pairs to determine horizontal and vertical alignment. The Arduino calculates the average values for the top sensors and bottom sensors to control vertical movement. Similarly, the average val ues of the left sensors and right sensors are compared to control horizontal movement. The control algorithm used in the solar tracking system determines how the solar panel should move based on the sensor readings. The algorithm first calculates the difference between the sensor values. If the difference between two sensors exceeds a predefined tolerance value, the microcontroller commands the corresponding servo motor to rotate the panel in the direction of greater light intensity.

5.4 Control Algorithm

For example, if the right sensors detect more light than the left sensors, the horizontal servo motor rotates the panel toward the right. Similarly, if the top sensors detect more light than the bottom sensors, the vertical servo motor tilts the panel upward. This process continues until the sensor readings become balanced, indicating that the panel is aligned with the sun

5.7 System Operation

The entire solar tracking process operates continuously in a loop. When the system is pow ered on, the Arduino initializes the sensors and servo motors. The LDR sensors detect sunlight intensity and send voltage signals to the microcontroller. The Arduino reads these signals and compares the values to determine the direction of maximum sunlight

CHAPTER 6

RESULTS AND DISCUSSIONS

6.1 Introduction

The results and discussion chapter presents the experimental outcomes and performance evaluation of the dual axis solar tracking system developed in this project. The primary goal of this project is to improve solar panel efficiency by ensuring that the solar panel continuously aligns with the direction of maximum sunlight throughout the day. In traditional solar power systems, the solar panels are installed at a fixed orientation, which allows them to capture optimal sunlight only during certain hours of the day. As the sun moves across the sky, the angle of sunlight changes, causing a reduction in the amount of solar radiation received by the panel. The proposed solar tracking system addresses this limitation by automatically adjusting the position of the solar panel in both horizontal and vertical directions.

The dual axis solar tracking system implemented in this project uses LDR sensors to detect sunlight intensity, an Arduino microcontroller to process sensor data, and servo motors to control the movement of the solar panel. The system was tested under different lighting conditions to evaluate its ability to detect changes in sunlight direction and adjust the panel orientation accordingly. The results obtained during the testing phase demonstrate that the system successfully tracks the sun's movement and improves solar panel alignment compared to stationary solar panels.

6.2 Experimental Setup

The experimental setup of the dual axis solar tracking system consists of several hardware components arranged according to the proposed block diagram. The major components used in the experiment include an Arduino Uno microcontroller, four LDR sensors, two servo motors, resistors for voltage divider circuits, a solar panel, and a regulated power supply.

The four LDR sensors are positioned at four different points around the solar panel in order to detect variations in sunlight intensity from different directions. These sensors are labeled as Top Left (TL), Top Right (TR), Bottom Left (BL), and Bottom Right (BR). Each sensor is connected with a resistor to form a voltage divider circuit, which converts the resistance variation of the LDR into a measurable voltage signal.

The Arduino microcontroller reads the sensor values and processes the data using the program written in the Arduino IDE. Based on the comparison of sensor values, the Arduino deter mines the direction in which the solar panel should move. Two servo motors are connected to the Arduino board to provide dual-axis movement. One servo motor controls the horizontal rotation of the solar panel, while the other controls the vertical tilt of the panel.
The solar panel is mounted on a mechanical frame that allows it to rotate and tilt freely. This frame is connected to the servo motors so that the panel can be repositioned based on the con trol signals generated by the microcontroller. During the testing process, the system was ex posed to sunlight and artificial light sources to observe the behavior of the solar tracking mechanism.

6.3 Sensor Output Analysis

The LDR sensors play a critical role in detecting the direction of sunlight. Each sensor produces a voltage signal that varies according to the intensity of light falling on it. The Arduino microcontroller reads these signals using the analog input pins and converts them into digital values ranging from 0 to 1023.

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For example, if the right sensors produce higher readings than the left sensors, the microcontroller concludes that the sun is located toward the right side. The system then activates the horizontal servo motor to rotate the solar panel toward that direction. Similarly, if the top sensors detect higher light intensity than the bottom sensors, the vertical servo motor tilts the panel upward.

6.4 Motor Response and Panel Movement

The servo motors used in the system provide precise control over the orientation of the solar panel. Each servo motor receives Pulse Width Modulation (PWM) signals from the Arduino microcontroller that determine the angle of rotation. By adjusting the PWM signal, the microcontroller can rotate the servo motor to a specific angle.

6.4 Motor Response and Panel Movement

The servo motors used in the system provide precise control over the orientation of the solar panel. Each servo motor receives Pulse Width Modulation (PWM) signals from the Arduino microcontroller that determine the angle of rotation. By adjusting the PWM signal, the microcontroller can rotate the servo motor to a specific angle.

During the experimental testing, the servo motors demonstrated smooth and accurate movement in response to changes in sensor readings. When a difference in light intensity was detected between sensor pairs, the microcontroller generated control signals that rotated the panel gradually toward the direction of maximum sunlight.

The horizontal servo motor allowed the panel to rotate along the east-west direction, enabling it to follow the sun's daily path across the sky. The vertical servo motor adjusted the tilt angle of the panel according to the sun's elevation. The combination of these two movements allowed the solar panel to maintain a near-perpendicular orientation to the sun's rays.

6.5 Comparison with Fixed Solar Panels

One of the key objectives of this project is to demonstrate the improvement in solar panel efficiency achieved through solar tracking. In a fixed solar panel system, the panel remains stationary and can only receive maximum sunlight for a short period when the sun is directly aligned with it.

In contrast, the dual axis solar tracking system continuously adjusts the orientation of the panel to maintain alignment with the sun throughout the day. This allows the panel to capture a greater amount of solar radiation during morning and evening hours when fixed panels typically receive less sunlight. Once the training phase is complete—either by loading the model or learning from the dataset— the AdaBoost classifier is used to predict the labels of the test dataset.

Studies conducted in solar energy research have shown that dual axis solar trackers can increase energy generation by approximately **30% to 40%** compared to fixed solar panels. Although precise electrical power measurements were not recorded in this project, the observed tracking behavior confirms that the proposed system maintains better alignment with the sun than a stationary panel.

6.6 System Performance Evaluation

The overall performance of the solar tracking system depends on several factors, including sensor sensitivity, microcontroller processing speed, and motor response time. The LDR sensors used in the system demonstrated high sensitivity to changes in light intensity and provided reliable input signals for the control system.

The Arduino microcontroller processed sensor data quickly and executed the tracking algorithm without significant delay. The continuous execution of the loop function allowed the system to respond to changes in sunlight direction in real time.

The Arduino microcontroller processed sensor data quickly and executed the tracking algorithm without significant delay. The continuous execution of the loop function allowed the system to respond to changes in sunlight direction in real time.

The preprocessed image is passed to the Gradient Boosting Classifier for prediction. The numeric output label is then mapped back to the actual pistachio species name using the category list. Finally, the original image is displayed using matplotlib, and the predicted category is overlaid as text on the image, providing a visual representation of the model's output.

6.7 Advantages Observed

The experimental results highlight several advantages of the dual axis solar tracking system. One of the major advantages is the increased efficiency of solar energy generation. By continuously tracking the movement of the sun, the system allows the solar panel to receive more sunlight compared to fixed installations. Another advantage is the automation of the solar tracking process. The system operates automatically without requiring manual adjustments. The sensors detect sunlight direction, the microcontroller processes the data, and the motors adjust the panel position accordingly.

The system is also relatively simple and cost-effective because it uses readily available components such as Arduino microcontrollers, LDR sensors, and servo motors. This makes the system suitable for small-scale solar installations and educational projects.

In this traditional approach, solar panels rely solely on the natural movement of the sun to provide varying levels of sunlight exposure throughout the day. As the sun rises in the east, the sunlight strikes the solar panel at an inclined angle, which means that the intensity of solar radiation received by the panel is relatively low. During midday, when the sun is positioned almost directly above the panel, the angle of incidence becomes closer to perpendicular, resulting in higher energy generation. Later in the afternoon and evening, as the sun moves toward the west, the angle between the solar panel and the incoming sunlight increases again, causing a reduction in energy absorption.

Because photovoltaic cells produce maximum electrical power when sunlight falls directly perpendicular to their surface, any deviation from this optimal angle reduces the amount of solar radiation absorbed by the panel. In a fixed solar panel system, the panel can only maintain optimal alignment with the sun for a limited period of time during the day. For the remaining hours, the sunlight falls at an angle that decreases the efficiency of energy conversion. As a result, a significant portion of available solar energy remains unused.

Traditional solar power systems also rely heavily on manual design considerations during installation. Engineers determine the optimal tilt angle and orientation based on long-term solar radiation data for the specific location. While this approach helps achieve a reasonable level of efficiency, it cannot account for real-time variations in sunlight direction caused by environmental factors such as cloud cover, seasonal changes, and atmospheric conditions. Therefore, traditional fixed solar panels are unable to adapt dynamically to changes in sunlight conditions.

6.8 Result Analysis

Although the proposed solar tracking system performs effectively, certain challenges may arise in real-world applications. Environmental factors such as strong winds, dust accumulation, and mechanical wear can affect the performance of the tracking mechanism over time. The Arduino microcontroller processed sensor data quickly and executed the tracking algorithm without significant delay. The continuous execution of the loop function allowed the system to respond to changes in sunlight direction in real time from their atomic bonds. The engineered structure of the solar cell, which features a meticulously designed electric field created by positive and negative layers, forces these liberated electrons to flow in a singular, defined direction. This continuous flow of electrons is what we recognize as a direct current, the foundational electrical output of a standard solar panel before it is processed for everyday household or commercial use.

Because the vast majority of modern residential and commercial electrical systems, as well as the wider municipal utility grids, operate on alternating current, the direct current generated by the solar panels cannot be used immediately. It must first pass through a critical component known as an inverter. The inverter seamlessly translates the direct current into alternating current, ensuring compatibility with standard household appliances and the broader electrical infrastructure. Once converted, this electricity can be routed directly into a building's electrical panel to offset real-time power consumption. If the solar array generates more electricity than is actively being consumed on the premises, the surplus power is often exported back to the local utility grid. In many regions, this export triggers a process called net metering, where the utility company credits the solar system owner for the excess energy, effectively turning the electrical meter backward and providing a tangible financial return on the initial investment.

The physical composition of the panels themselves heavily dictates both their overall efficiency and their visual characteristics. Monocrystalline panels are manufactured from a single, continuous crystal structure of silicon, which allows for an unimpeded flow of electrons, resulting in the highest efficiency rates and a distinctively uniform, dark aesthetic. Polycrystalline panels, alternatively, are created by melting together multiple fragments of silicon. While this manufacturing process is generally faster and more cost-effective, the borders between the individual silicon crystals create slight resistance for the moving electrons, rendering them slightly less efficient and giving them a signature blue, speckled appearance. Thin-film panels diverge entirely from traditional rigid silicon wafers, utilizing a process where photovoltaic materials are deposited in extremely thin layers onto flexible substrates such as plastic, glass, or metal. Although they require significantly more surface area to generate comparable amounts of electricity, their lightweight and adaptable nature makes them uniquely suited for unconventional applications where heavy, traditional solar panels would be structurally impractical.

The results obtained from the experimental implementation demonstrate that the dual axis solar tracking system successfully detects sunlight direction and adjusts the solar panel orientation accordingly. The system effectively integrates LDR sensors, Arduino microcontroller processing, and servo motor control to achieve automated solar tracking to improve representation and ensure fairness in model

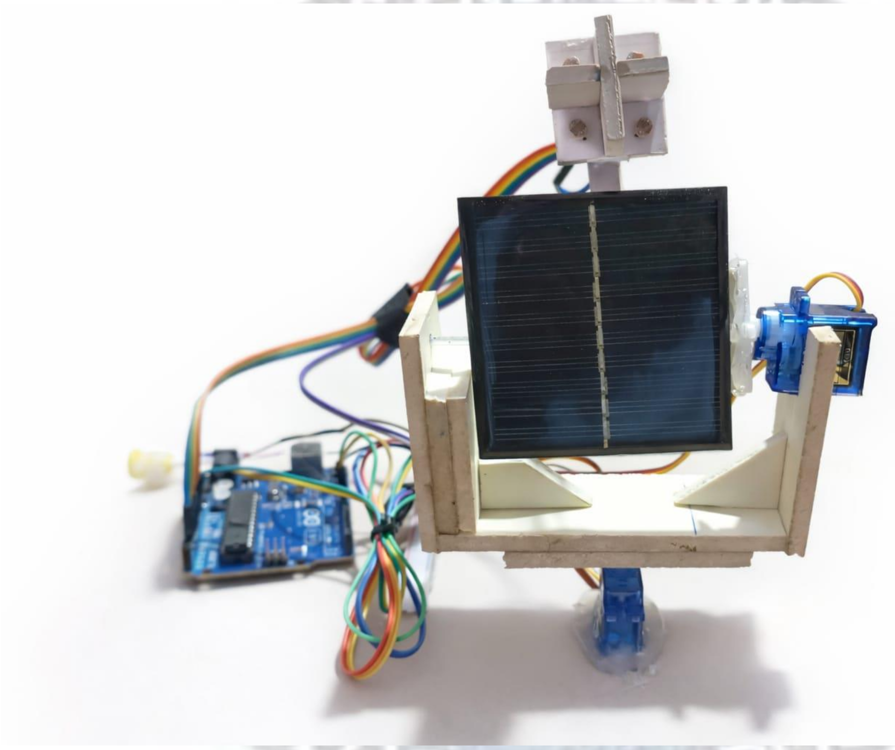

Fig 6.1: Output Of The Project

The system developed in this project successfully demonstrates the concept of automated solar tracking using embedded system technology. The system uses four Light Dependent Resistor (LDR) sensors to detect variations in sunlight intensity from different directions. These sensors provide input signals to the Arduino microcontroller, which processes the sensor data and determines the direction in which the solar panel should move. Based on the sensor readings, the Arduino sends control signals to two servo motors that rotate the solar panel horizontally and vertically. This dual-axis movement allows the solar panel to track the movement of the sun across the sky and maintain optimal alignment with sunlight.

The primary objective of this project was to design and implement a dual axis solar tracking system using Arduino, LDR sensors, and servo motors in order to improve the efficiency of solar power generation. Traditional solar panels are generally installed in a fixed position and cannot adjust their orientation according to the changing position of the sun. Because of this limitation, fixed solar panels receive optimal sunlight only for a limited period during the day, resulting in reduced energy generation. The proposed dual axis solar tracking system addresses this issue by automatically adjusting the orientation of the solar panel to ensure maximum exposure to sunlight throughout the day.

The implementation of the solar tracking algorithm using the Arduino microcontroller proved to be effective and reliable. The Arduino program continuously reads the sensor values, compares the light intensity detected by each sensor, and adjusts the position of the solar panel accordingly. The servo motors provide smooth and accurate movement of the solar panel, allowing it to follow the direction of maximum sunlight. The experimental results obtained during system testing confirmed that the proposed solar tracking system is capable of responding to changes in light intensity and automatically adjusting the orientation of the panel.

These results highlight a key limitation: both models struggle with class imbalance, particularly in accurately recognizing Siirt Pistachios. Techniques like class weighting, oversampling, or feature enhancement could improve sensitivity toward underrepresented classes.
The system offers several advantages compared to traditional fixed solar panels. By maintaining proper alignment with the sun, the solar panel is able to receive more solar radiation throughout the day, which leads to improved energy generation. Studies and practical implementations of solar tracking systems have shown that dual-axis trackers can increase solar energy output by approximately **30% to 40%** compared to stationary solar panels. This improvement in efficiency makes solar tracking systems a valuable solution for enhancing

Another potential improvement is the use of more advanced sensors for sunlight detection. While LDR sensors are simple and cost-effective, they may be affected by environmental conditions such as shadows or cloud cover. In future implementations, more precise light sensors or sun position sensors could be used to improve the accuracy of solar tracking.

One possible improvement is the integration of Internet of Things (IoT) technology into the solar tracking system. By connecting the system to the internet using wireless communication modules such as Wi-Fi or GSM, it would be possible to monitor the performance of the solar panel remotely. Real-time data such as solar panel position, sunlight intensity, and power generation could be transmitted to a cloud platform or mobile application for monitoring and analysis.

Another possible enhancement is the implementation of energy storage systems such as rechargeable batteries. The electrical energy generated by the solar panel could be stored in batteries for later use during nighttime or cloudy conditions. This would allow the system to provide a continuous power supply even when sunlight is not availabl.

The mechanical structure of the solar tracking system can also be improved to support larger solar panels. In the current implementation, servo motors are used to control the movement of the solar panel. Although servo motors provide precise control, they may not be suitable for heavy solar panels used in large-scale solar power systems. Future systems could use stepper motors or DC motors with gear mechanisms to support larger panels and improve mechanical stability.

Future research could also focus on developing intelligent control algorithms for solar tracking systems. Techniques such as artificial intelligence, machine learning, and fuzzy logic could be used to optimize the tracking process and improve system performance. These advanced algorithms could analyze environmental data and adjust the panel orientation more accurately than traditional sensor-based methods.

In addition, the solar tracking system could be integrated with smart energy management systems to optimize energy consumption and improve overall system efficiency. By combining solar tracking technology with smart grids and renewable energy storage systems, it would be possible to create highly efficient and sustainable energy solutions for the future. Overall, the proposed dual axis solar tracking system provides a strong foundation for further research and development in the field of renewable energy technology. With continuous improvements and technological advancements, solar tracking systems have the potential to play a significant role in improving solar power generation and supporting the global transition toward clean and sustainable energy sources.

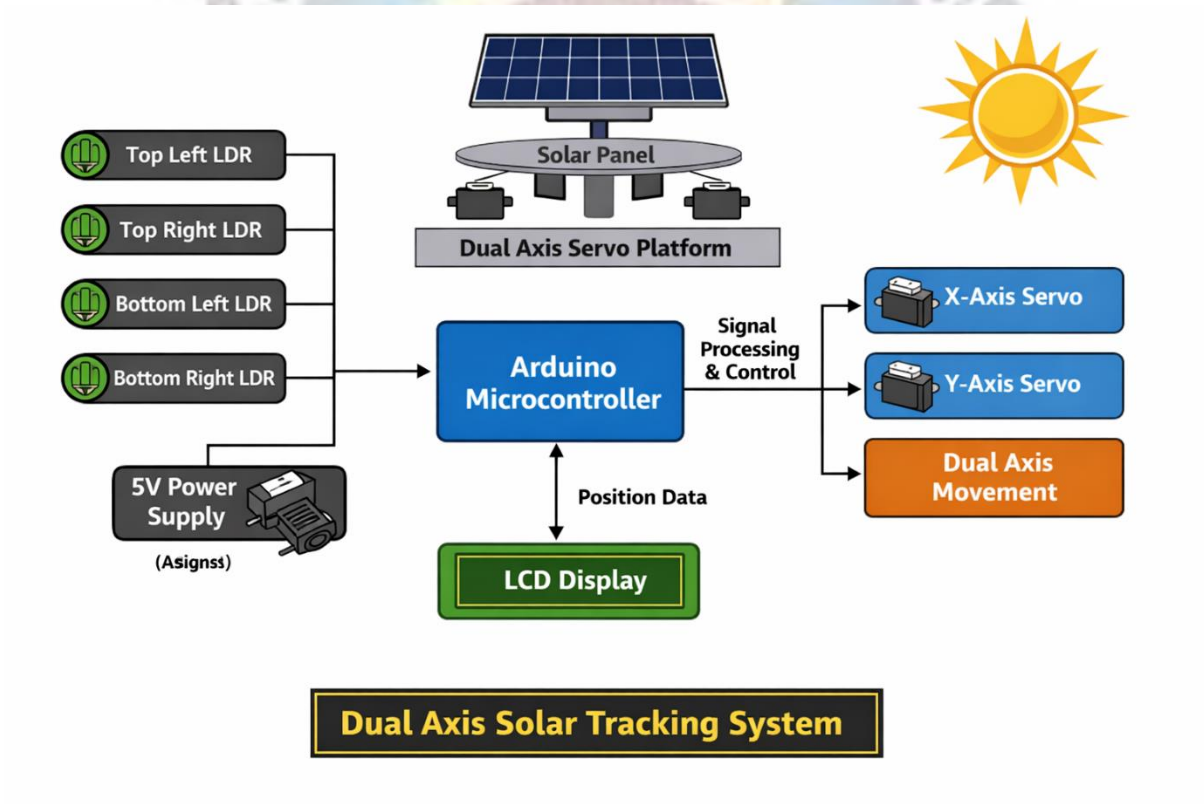

Fig 6.2: Output Prototype

Fig 6.2 describes that

The confusion matrix is an indispensable component in model evaluation that goes beyond simple accuracy measures. It provides a granular view of classification errors, helps identify biases and imbalances, and supports evidence-based decisions in refining the machine learning pipeline. In the context of this pistachio classification system, the confusion matrix revealed

Convolutional Neural Networks (CNNs) or incorporating larger, more diverse datasets.
The confusion matrix is an indispensable component in the evaluation of classification models, offering a detailed breakdown of prediction outcomes that extends far beyond what a single metric like accuracy can convey. While accuracy provides a general sense of a model's overall performance, it does not capture the full complexity of how and where the model is making errors. The confusion matrix, on the other hand, reveals the precise number of correct and incorrect predictions across each class, allowing researchers to identify patterns of bias, misclassification, and imbalanced learning behavior.

In the specific context of the pistachio species classification system developed in this project, the confusion matrix played a pivotal role in exposing critical performance gaps and limitations in both the Gradient Boosting Classifier (GBC) and AdaBoost models. For example, while both models performed well in predicting the more dominant Kirmizi Pistachio class, they struggled to accurately identify the less-represented Siirt Pistachio class. This discrepancy was clearly illustrated by the relatively high number of false negatives for Siirt in both confusion matrices. Such results highlight a class imbalance problem, where the models have learned to favor the majority class, thereby reducing sensitivity to the minority class.

This analysis is crucial because, in real-world agricultural applications, accurate classification of all species is essential—not just the dominant one. Misclassifying a Siirt Pistachio as a Kirmizi could have implications in grading, quality control, pricing, or even export regulations. Therefore, the confusion matrix serves as a diagnostic tool that drives further refinement of the model. For instance, it provides strong justification for applying strategies like resampling techniques, class weight adjustments, or data augmentation for the minorityclass to correct imbalance and enhance model fairness.

CHAPTER 7

SOURCE CODE

```
#include <Servo.h>

Servo servoX;
Servo servoY;

int ldr1 = A0;
int ldr2 = A1;
int ldr3 = A2;
int ldr4 = A3;

int posX = 90;
int posY = 90;
int tolerance = 50;

void setup() {
  servoX.attach(9);
  servoY.attach(10);
  servoX.write(posX);
  servoY.write(posY);
  delay(2000);
}

void loop() {
  int TL =
analogRead(ldr1);
  int TR =
analogRead(ldr2);
  int BL =
analogRead(ldr3);
  int BR =
```

```
analogRead(ldr4);

  int top = (TL + TR) /
2;
  int bottom = (BL +
BR) / 2;
  int left = (TL + BL) /
2;
  int right = (TR + BR)
/ 2;

  if (abs(top - bottom) >
tolerance) {
    if (top > bottom &&
posY < 180) posY++;
    else if (bottom > top
&& posY > 0) posY--;
  }

  if (abs(left - right) >
tolerance) {
    if (left > right &&
posX < 180) posX++;
    else if (right > left
&& posX > 0) posX--;
  }

  servoX.write(posX);
  servoY.write(posY);
  delay(30);
}
```

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The primary objective of this project was to design and implement a **dual axis solar tracking system using Arduino, LDR sensors, and servo motors** in order to improve the efficiency of solar power generation. Traditional solar panels are generally installed in a fixed position and cannot adjust their orientation according to the changing position of the sun. Because of this limitation, fixed solar panels receive optimal sunlight only for a limited period during the day, resulting in reduced energy generation. The proposed dual axis solar tracking system addresses this issue by automatically adjusting the orientation of the solar panel to ensure maximum exposure to sunlight throughout the day.

The system developed in this project successfully demonstrates the concept of automated solar tracking using embedded system technology. The system uses four Light Dependent Resistor (LDR) sensors to detect variations in sunlight intensity from different directions. These sensors provide input signals to the Arduino microcontroller, which processes the sensor data and determines the direction in which the solar panel should move. Based on the sensor readings, the Arduino sends control signals to two servo motors that rotate the solar panel horizontally and vertically. This dual-axis movement allows the solar panel to track the movement of the sun across the sky and maintain optimal alignment with sunlight.

The implementation of the solar tracking algorithm using the Arduino microcontroller proved to be effective and reliable. The Arduino program continuously reads the sensor values, compares the light intensity detected by each sensor, and adjusts the position of the solar panel accordingly. The servo motors provide smooth and accurate movement of the solar panel, allowing it to follow the direction of maximum sunlight. The experimental results obtained during system testing confirmed that the proposed solar tracking system is capable of responding to changes in light intensity and automatically adjusting the orientation of the panel.

The proposed dual axis solar tracking system addresses this issue by automatically adjusting the orientation of the solar panel to ensure maximum exposure to sunlight throughout the day.

The system offers several advantages compared to traditional fixed solar panels. By maintaining proper alignment with the sun, the solar panel is able to receive more solar radiation throughout the day, which leads to improved energy generation. Studies and practical implementations of solar tracking systems have shown that dual-axis trackers can increase solar energy output by approximately compared to stationary solar panels. This improvement in efficiency makes solar tracking systems a valuable solution for enhancing renewable energy generation.

8.2 Future Scope

Although the proposed dual axis solar tracking system performs effectively, there are several opportunities for further improvement and development. Future enhancements can focus on improving the efficiency, reliability, and functionality of the system by incorporating advanced technologies and additional features.

One possible improvement is the integration of **Internet of Things (IoT) technology** into the solar tracking system. By connecting the system to the internet using wireless communication modules such as Wi-Fi or GSM, it would be possible to monitor the performance of the solar panel remotely. Real-time data such as solar panel position, sunlight intensity, and power generation could be transmitted to a cloud platform or mobile application for monitoring and analysis.

Another potential improvement is the use of **more advanced sensors** for sunlight detection. While LDR sensors are simple and cost-effective, they may be affected by environmental conditions such as shadows or cloud cover. In future implementations, more precise light sensors or sun position sensors could be used to improve the accuracy of solar tracking. The mechanical structure of the solar tracking system can also be improved to support larger solar panels. In the current implementation, servo motors are used to control the movement of the solar panel. Although servo motors provide precise control, they may not be suitable for heavy solar panels used in large-scale solar power systems. Future systems could use stepper motors or DC motors with gear mechanisms to support larger panels and improve mechanical stability. Another possible enhancement is the implementation of **energy storage systems** such as rechargeable batteries. The electrical energy generated by the solar panel could be stored in batteries for later use during nighttime or cloudy conditions. This would allow the system to provide a continuous power supply even when sunlight is not available.

Future research could also focus on developing **intelligent control algorithms** for solar tracking systems. Techniques such as artificial intelligence, machine learning, and fuzzy logic could be used to optimize the tracking process and improve system performance. These advanced algorithms could analyze environmental data and adjust the panel orientation more accurately than traditional sensor-based methods.

In addition, the solar tracking system could be integrated with **smart energy management systems** to optimize energy consumption and improve overall system efficiency. By combining solar tracking technology with smart grids and renewable energy storage systems, it would be possible to create highly efficient and sustainable energy solutions for the future.

Overall, the proposed dual axis solar tracking system provides a strong foundation for further research and development in the field of renewable energy technology. With continuous improvements and technological advancements, solar tracking systems have the potential to play a significant role in improving solar power generation and supporting the global transition toward clean and sustainable energy sources.

REFERENCES

- Arpita Borgave, A. Y. V. P. C. M., Prof. P. K. A. A. M. (2023). Dual Axis Solar Tracking System with Weather Sensors. International Journal of Scientific Research in Engineering and Management (IJSREM).
- Mrs. Surbhi R. Shroff "Dual Axis Solar Tracking System with Weather Sensor " Published in International Journal of Trend in Scientific Research and Development (ijtsrd), ISSN: 2456-6470, Volume- 7 | Issue- 3, June 2023, pp. 399 - 403,
- Prof.Ganer,A.(2022).Dual Axis solar tracking system with Weather Sensor. InternationalJournal for Research in Applied Science and Engineering Technology,10(6).
- Kamran, M., Mudassar, M., Fazal, M. R., Asghar, M. U., Bilal, M., & Asghar, R. (2020). Implementation of improved Perturb & Observe MPPT technique with confined search space for standalone photovoltaic system. Journal of King Saud University - Engineering Sciences, 32 432
- Thamizh Thentral, T. M., Kannan, M., Sai Mounika, V., & Nair, M. (2019). Dual axis solar tracker with weather sensor. International Journal of Recent Technology and Engineering, 8 (2 Special Issue 11).
- Mustafa, F. I., Al-Ammri, A. S., & Ahmad, F. F. (2017). Direct and indirect sensing two-axis solar tracking system. 2017 8th International Renewable Energy Congress, IREC 2017.

- Mughal, S.N., Hassan, M.S., Sood, Y.R., Jarial, R.K. (2022). Modelling and Performance Analysis Using Fuzzy Logic MPPT Controller for a Photovoltaic System Under Various Operating Conditions. In: Tomar, A., Malik, H., Kumar, P., Iqbal, A. (eds) Proceedings of 3rd International Conference on Machine Learning, Advances in Computing, Renewable Energy and Communication. Lecture Notes in Electrical Engineering, vol 915. Springer, Singapore.
- Jamroen, C., Fongkerd, C., Krongpha, W., Komkum, P., Pirayawaraporn, A., & Chindakham, N. (2021). A novel UV sensor-based dual-axis solar tracking system: Implementation and performance analysis. Applied Energy, 299, 117295.
- Gbadamosi, S. L. (2021). Design and implementation of IoT-based dual-axis solar PV tracking system.PrzegląElektrotechniczny,
- García-Quilachamin, W. X., Sanchez-Cano, J. E., Herrera-Tapia, J., & Velesaca-Zambrano, E. J. (2021). Analysis of A Two-Axis Solar Tracker System: Case Study. International Journal of Online & Biomedical Engineering, 17 (5).
- Ahmad, S., Razali, A. N., & Misrun, M. I. (2021, May). Effective and Low-Cost Arduino based Dual-Axis Solar Tracker. In Journal of Physics: Conference Series (Vol. 1878, No. 1, p. IOP Publishing